

Here's a clean, self-contained description of the **GenAI-powered clinical NLP concept schema** you can drop into a document and export to PDF.

1. Overview

Goal: Represent **clinical concepts extracted from unstructured notes** by GenAI-powered clinical NLP, in a **FHIR-aligned but simple relational schema**, without medallion language or raw FHIR JSON.

Key design points:

- Separate **note context** from **clinical concepts**.
 - Store **normalized clinical entities** (problems, labs, meds, procedures, allergies, social/family, care plan, oncology).
 - Capture **text provenance** and **context attributes** (negation, temporality, certainty, severity, anatomy, dosage, stage, etc.).
 - Optionally normalize codes via a small **terminology dimension**.
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2. Note / Context Tables

2.1 NOTE_DOCUMENT

Purpose: One row per clinical note processed by GenAI-powered clinical NLP.

Columns (core):

- `document_id` (PK)
- `patient_id`
- `encounter_id` (nullable)
- `note_type`
 - Examples: `H_AND_P`, `PROGRESS`, `DISCHARGE_SUMMARY`, `RADIOLOGY_REPORT`, `PATHOLOGY_REPORT`, `NURSING_NOTE`, `ED_NOTE`, `CONSULT_NOTE`, etc.
- `source_system` (e.g., "Epic", "Cerner", "HIE_X", etc.)
- `author` (free text or provider ID)
- `created_datetime` (note creation/signed time)
- `raw_text` (full note body, TEXT or VARIANT / pointer)

2.2 NOTE_SECTION (optional but recommended)

Purpose: Represent logical sections inside a note (useful for precision and QA).

Columns:

- `section_id` (PK)
- `document_id` (FK → NOTE_DOCUMENT)
- `section_name`
 - Examples: CHIEF_COMPLAINT, HPI, PMH, PSH, MEDICATIONS, ALLERGIES, SOCIAL_HISTORY, FAMILY_HISTORY, ROS, PHYSICAL_EXAM, LABS, IMAGING, ASSESSMENT, PLAN, IMPRESSION, etc.
- `start_offset` (INT; character index in `raw_text`)
- `end_offset` (INT)

(PATIENT and ENCOUNTER tables are assumed to exist already and are not redefined here.)

3. Core Clinical Concept Tables (FHIR-Aligned, Relational)

3.1 CONDITION

Purpose: Diagnoses, problems, some “problem-worthy” symptoms, risk states.

Columns:

- `condition_id` (PK)
- `patient_id`
- `encounter_id` (nullable)
- `code` (e.g., ICD-10 or SNOMED code)
- `code_system` (e.g., ICD10CM, SNOMED_CT)
- `display` (human-readable label)
- `clinical_status`
 - active, recurrence, resolved, inactive, unknown
- `verification_status`
 - provisional, differential, confirmed, refuted, entered_in_error
- `category`
 - Examples: PROBLEM_LIST_ITEM, ENCOUNTER_DIAGNOSIS, SYMPTOM, RISK_FACTOR, HISTORY_OF
- `onset_datetime` (nullable)
- `abatement_datetime` (nullable)

- `severity_code` (nullable)
 - `severity_display` (nullable; e.g., “mild”, “moderate”, “severe”)
 - `body_site_code` (nullable)
 - `body_site_display` (nullable)
 - `laterality` (nullable; e.g., `LEFT`, `RIGHT`, `BILATERAL`, `UNKNOWN`)
 - `recorded_date` (nullable)
 - `source`
 - `EHR_STRUCTURED`, `GENAI_NLP_NOTE`, `OTHER`
 - `provenance_document_id` (nullable, FK → `NOTE_DOCUMENT`)
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3.2 OBSERVATION

Purpose: Labs, vitals, exam findings, imaging measurements, scores, certain social-history facts.

Columns:

- `observation_id` (PK)
- `patient_id`
- `encounter_id` (nullable)
- `code`
- `code_system` (e.g., `LOINC`, `SNOMED_CT`, custom)
- `display`
- `category`
 - `VITAL_SIGNS`, `LAB`, `IMAGING`, `EXAM`, `SCORE`, `SOCIAL_HISTORY`, `OTHER`
- `status`
 - `REGISTERED`, `PRELIMINARY`, `FINAL`, `AMENDED`, `CANCELLED`
- `effective_datetime`
- Value fields (use whichever apply):
 - `value_quantity` (NUMBER, nullable)
 - `value_unit` (nullable)
 - `value_code` (nullable)
 - `value_code_system` (nullable)
 - `value_display` (nullable)
 - `value_string` (nullable)
- `interpretation` (nullable; e.g., `HIGH`, `LOW`, `ABNORMAL`, `NORMAL`, `CRITICAL_HIGH`, `CRITICAL_LOW`)
- `body_site_code` (nullable)
- `body_site_display` (nullable)
- `laterality` (nullable)
- `method` (nullable)

- `source` (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - `provenance_document_id` (nullable, FK → NOTE_DOCUMENT)
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3.3 PROCEDURE

Purpose: Surgical, diagnostic, therapeutic, and imaging procedures.

Columns:

- `procedure_id` (PK)
 - `patient_id`
 - `encounter_id`
 - `code`
 - `code_system` (e.g., SNOMED_CT, CPT, ICD10PCS, local)
 - `display`
 - `status`
 - PREPARATION, IN_PROGRESS, COMPLETED, NOT_DONE, STOPPED, UNKNOWN
 - `category` (e.g., SURGICAL, DIAGNOSTIC, IMAGING, THERAPEUTIC, OTHER)
 - `performed_start` (nullable)
 - `performed_end` (nullable)
 - `reason_condition_id` (nullable, FK → CONDITION)
 - `body_site_code` (nullable)
 - `body_site_display` (nullable)
 - `laterality` (nullable)
 - `source` (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - `provenance_document_id` (nullable, FK → NOTE_DOCUMENT)
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3.4 MEDICATION_REQUEST

Purpose: Medication orders / prescriptions; often from meds lists, discharge summaries, plans.

Columns:

- `medication_request_id` (PK)
- `patient_id`
- `encounter_id` (nullable)
- `medication_code`
- `medication_system` (e.g., RXNORM, SNOMED_CT, local)
- `medication_display`

- status
 - ACTIVE, COMPLETED, STOPPED, CANCELLED, DRAFT, UNKNOWN
 - intent (e.g., ORDER, PLAN, PROPOSAL)
 - dosage_text (free-text sig)
 - dose (NUMBER, nullable)
 - dose_unit (nullable)
 - route_code (nullable)
 - route_display (nullable)
 - frequency_text (e.g., “BID”, “q6h PRN”)
 - frequency_norm (nullable normalized representation)
 - duration_value (nullable)
 - duration_unit (e.g., DAYS, WEEKS, MONTHS)
 - authored_on (nullable)
 - requester (nullable; prescriber)
 - indication_condition_id (nullable, FK → CONDITION)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, FK → NOTE_DOCUMENT)
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3.5 ALLERGY_INTOLERANCE

Purpose: Drug/food/environmental allergies and intolerances.

Columns:

- allergy_id (PK)
 - patient_id
 - substance_code
 - substance_system (e.g., RXNORM, SNOMED_CT)
 - substance_display
 - reaction_code (nullable)
 - reaction_display (nullable; e.g., “rash”, “anaphylaxis”)
 - severity (e.g., MILD, MODERATE, SEVERE, LIFE_THREATENING, UNKNOWN)
 - criticality (e.g., LOW, HIGH, UNABLE_TO_ASSESS, UNKNOWN)
 - verification_status (e.g., UNCONFIRMED, CONFIRMED, REFUTED, ENTERED_IN_ERROR)
 - onset_datetime (nullable)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, FK → NOTE_DOCUMENT)
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3.6 ADVERSE_EVENT

Purpose: Unintended harmful clinical events (drug/device/procedure related).

Columns:

- adverse_event_id (PK)
 - patient_id
 - encounter_id (nullable)
 - event_code
 - event_system (e.g., MEDDRA, SNOMED_CT)
 - event_display
 - seriousness (SERIOUS, NON_SERIOUS, UNKNOWN)
 - severity (MILD, MODERATE, SEVERE, LIFE_THREATENING)
 - onset_datetime
 - resolution_datetime (nullable)
 - outcome (e.g., RECOVERED, RECOVERING, FATAL, UNKNOWN, OTHER)
 - suspect_medication_request_id (nullable, FK → MEDICATION_REQUEST)
 - suspect_device_id (nullable, if a DEVICE table exists)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, FK → NOTE_DOCUMENT)
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3.7 SOCIAL_HISTORY_OBSERVATION

Purpose: Tobacco, alcohol, drugs, occupation, housing, social context.

Columns:

- social_history_id (PK)
- patient_id
- code
- code_system (e.g., SNOMED_CT, LOINC)
- display
- value_string (e.g., “1 ppd × 30 yrs”, “drinks 2–3 beers nightly”)
- value_quantity (nullable; numeric form like pack-years)
- value_unit (nullable)
- effective_period_start (nullable)
- effective_period_end (nullable)
- source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
- provenance_document_id (nullable, FK → NOTE_DOCUMENT)

3.8 FAMILY_MEMBER_HISTORY

Purpose: Conditions affecting relatives (risk context).

Columns:

- family_history_id (PK)
 - patient_id
 - relationship_code (e.g., MOTHER, FATHER, SIBLING, CHILD, OTHER)
 - relationship_display
 - condition_code
 - condition_system
 - condition_display
 - onset_age (nullable)
 - deceased_flag (nullable)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, FK → NOTE_DOCUMENT)
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3.9 CARE_PLAN_ITEM

Purpose: Goals, actions, referrals, follow-ups defined in the plan.

Columns:

- care_plan_item_id (PK)
 - patient_id
 - encounter_id (nullable)
 - item_type
 - GOAL, ACTION, REFERRAL, FOLLOW_UP, EDUCATION, MONITORING
 - description (free text)
 - target_condition_id (nullable, FK → CONDITION)
 - due_datetime (nullable)
 - status (PLANNED, IN_PROGRESS, COMPLETED, CANCELLED, UNKNOWN)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, FK → NOTE_DOCUMENT)
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3.10 TUMOR_EPISODE (Optional, Oncology-Focused)

Purpose: Consolidated tumor episode information: stage, grade, histology, response.

Columns:

- tumor_episode_id (PK)
 - patient_id
 - primary_condition_id (FK → CONDITION, primary cancer)
 - primary_site_code
 - primary_site_system
 - primary_site_display
 - stage_group (e.g., "IIB", "IVB")
 - tnm_t (nullable)
 - tnm_n (nullable)
 - tnm_m (nullable)
 - grade (e.g., "G3", "Gleason 4+3=7")
 - histology_code (nullable)
 - histology_system (nullable)
 - histology_display (nullable)
 - performance_status_scale (e.g., ECOG, KARNOFSKY)
 - performance_status_value (e.g., 0–4, 0–100)
 - response_status (CR, PR, SD, PD, UNKNOWN)
 - source (EHR_STRUCTURED, GENAI_NLP_NOTE, OTHER)
 - provenance_document_id (nullable, typically pathology/radiology/oncology notes)
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4. GenAI-Powered Clinical NLP Layer

4.1 NLP_NOTE_ENTITY_MENTION

Purpose: One row per mention (span) that GenAI/NLP extracts from free text.

Columns:

- mention_id (PK)
- document_id (FK → NOTE_DOCUMENT)
- section_id (nullable, FK → NOTE_SECTION)
- patient_id
- encounter_id (nullable)
- start_offset (INT)

- `end_offset` (INT)
 - `text` (exact substring from `raw_text`)
 - `candidate_type`
 - Examples: `CONDITION`, `SYMPTOM`, `MEDICATION`, `LAB_TEST`, `PROCEDURE`, `ANATOMY`, `SOCIAL_HISTORY`, `FAMILY_HISTORY`, `ALLERGY`, `ADVERSE_EVENT`, `CARE_PLAN`, `TUMOR`, `OUTCOME`, `RISK_FACTOR`, etc.
 - **Normalization output:**
 - `norm_resource_type`
 - `CONDITION`, `OBSERVATION`, `PROCEDURE`, `MEDICATION_REQUEST`, `ALLERGY_INTOLERANCE`, `ADVERSE_EVENT`, `SOCIAL_HISTORY_OBSERVATION`, `FAMILY_MEMBER_HISTORY`, `CARE_PLAN_ITEM`, `TUMOR_EPISODE`
 - `norm_resource_id` (nullable FK to the chosen table)
 - `norm_code` (nullable)
 - `norm_code_system` (nullable)
 - `norm_display` (nullable)
 - `engine_family` (`LLM`, `RULES`, `HYBRID`)
 - `engine_name` (e.g., `claude-3-5-sonnet`, `internal-ner-v2`)
 - `engine_version`
 - `confidence_score` (0–1)
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4.2 NLP_NOTE_ENTITY_ATTRIBUTE

Purpose: Attach context attributes to each mention (negation, temporality, etc.).

Columns:

- `mention_id` (FK → `NLP_NOTE_ENTITY_MENTION`)
 - `attribute_type`
 - `NEGATION`, `TEMPORALITY`, `CERTAINTY`, `EXPERIENCER`, `SEVERITY`, `BODY_SITE`, `LATERALITY`, `DOSAGE`, `ROUTE`, `FREQUENCY`, `DURATION`, `STAGE`, `GRADE`, `PERFORMANCE_STATUS`, `COURSE`, `HYPOTHETICAL_FLAG`
 - `value_text` (e.g., “negated”, “probable”, “left”, “acute on chronic”)
 - `value_code` (nullable; if mapped to SNOMED/etc.)
 - `value_system` (nullable)
 - `confidence_score` (0–1)
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4.3 NLP_NOTE_ENTITY_RELATION

Purpose: Relations between mentions (and, by extension, between normalized entities).

Columns:

- `relation_id` (PK)
- `subject_mention_id` (FK → NLP_NOTE_ENTITY_MENTION)
- `relation_type`
 - Examples: HAS_ANATOMICAL_SITE, INDICATION_OF, CAUSED_BY, COMPLICATES, FAMILY_HISTORY_OF, PART_OF_REGIMEN, HAS_STAGE, HAS_GRADE, ASSOCIATED_WITH
- `object_mention_id` (FK → NLP_NOTE_ENTITY_MENTION)
- `confidence_score` (0–1)

(If desired, you can add optional columns for `subject_resource_type/id` and `object_resource_type/id` after normalization.)

5. Optional Terminology Dimension

5.1 CODE_SYSTEM

- `code_system_id` (PK)
- `uri` (e.g., <http://snomed.info/sct>, <http://loinc.org>, <http://www.nlm.nih.gov/research/umls/rxnorm>)
- `name`

5.2 CONCEPT_DIMENSION

- `concept_id` (PK)
- `code_system_id` (FK → CODE_SYSTEM)
- `code`
- `display`
- `semantic_group`
 - DISEASE, SYMPTOM, PROCEDURE, LAB, MEDICATION, ANATOMY, DEVICE, SOCIAL, FAMILY, TUMOR, SCORE, OTHER

Clinical tables can either store (`code`, `code_system`) directly or reference `concept_id`.

This is the finalized **concept schema** for GenAI-powered clinical NLP over clinical notes, with:

- No medallion architecture references.

- No raw FHIR JSON tables.
- FHIR-aligned **relational** entities, plus a clean NLP layer and optional terminology layer.

You can paste this into a doc, add branding/diagrams, and export to PDF.

Do you want Snowflake CREATE TABLE DDL or an ER diagram (PNG/SVG) to include in the PDF?